

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed **small-scale protein purification studies** using AKTA systems and chromatography, improving process yield by **15%** for biopharmaceutical development.
- Authored comprehensive **SOPs, risk assessments, and technical reports** for laboratory processes, ensuring adherence to **GMP and ICH guidelines** and reducing documentation review time by **20%**.
- Supported root cause analysis investigations and troubleshooting of manufacturing issues, resolving **95%** of processing problems within SLA targets through systematic data analysis.
- Developed and maintained **small-scale models** for protein purification techniques, enhancing expertise in chromatography resin screening and filtration methodologies for process improvement.
- Managed cross-functional project plans involving **Process Development, Manufacturing, and Quality Assurance** teams, ensuring timely delivery of scientific outputs and fostering collaboration.
- Contributed to a culture of **Operational Excellence** by implementing standard work and 5S principles, increasing laboratory efficiency by **10%**.

SKILLS

Bioprocessing Techniques – Protein Purification, AKTA Systems, Chromatography, Resin Screening, Filtration, TFF.

Quality & Compliance – GMP, ICH Guidelines, SOPs, Risk Assessments, GLP.

Data Analysis & Reporting – Technical Reports, Data Integrity, Scientific Conclusions, Root Cause Analysis.

Laboratory Operations – Small-Scale Models, Process Development, Troubleshooting, Operational Excellence, 5S.

Project Management – Cross-functional Collaboration, Project Planning, Stakeholder Communication, Technical Agenda.

Scientific Expertise – Biochemistry, Biotechnology, Process Engineering, Biopharmaceutical Development.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Biologics Process Development & Optimization Platform (HCL Technologies)

- Developed and validated a suite of **Python scripts** to automate data collection and initial analysis from AKTA chromatography systems, reducing manual data processing time by **40%**.
- Designed and implemented experimental protocols for **small-scale protein purification** of therapeutic enzymes, achieving a **98% purity** target and improving downstream yield by **18%**.

- Created interactive **Power BI dashboards** to visualize purification process parameters, identify critical control points, and support troubleshooting efforts for manufacturing teams, leading to a **12% reduction** in batch deviations.

Financial Services Transaction Monitoring Dashboard (HCL Technologies)

- Enhanced a **payments operations dashboard** by implementing real-time monitoring for transaction flows, enabling proactive alerting for failed payments and reducing critical incident resolution time by **25%**.
- Developed automated **Python scripts** for data validation and reconciliation workflows, ensuring data accuracy and compliance with regulatory requirements for financial transactions.
- Supported the production environment with strict SLAs, focusing on operational alerting and audit trail maintenance for critical financial services operations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)